TECHNICAL DOCUMENTATION

APPLIED AI · RAG

2026 EDITION

Riansares 4R — Office, Cave, Bathroom and Client Screening Cases

REFERENCE DOCUMENTATION FOR THE AI ASSISTANT

RIANSARES 4R

Tarancón · 16400 · Cuenca · Spain

Documentation & AI: Germán Caballero

riansares.com **athagon.tech**

Riansares 4R — Office, Cave, Bathroom and Client Screening Cases

11. Case F — Office and Corporate Restroom Renovation

11.1. Context

Local corporate project divided into three packages:

1. Redistributing conference room and office space;
2. Comprehensive restroom renovation;
3. General painting and false ceiling.

The main lesson was that a corporate renovation should not be budgeted as a simple domestic project. There is coordination, documentation, parallel teams, compliance, schedule risk and visual quality.

11.2. Layout Changes

Works studied:

- Demolition of masonry partition;
- Demolition of built-in masonry shelving;
- Removal of debris;
- New acoustic partition;
- Conservation and selective closure of doors;
- Integration of flooring;
- Cat 6 data;
- Relocation of projector;
- Review of climate control;
- Repair of false ceiling in the affected area.

New partition:

- Metal framing;
- Mineral wool;
- Double boarding on both sides;
- Bands and sealing;
- Finish for painting.

Use of polyurethane foam as primary acoustic insulation was discarded.

11.3. Flooring

Criteria:

- Preserve usable laminate;
- Cut along partition line;
- Resolve unevenness with profiles;
- Install similar laminate, not necessarily identical;
- Do not promise invisible continuity when materials differ.

11.4. Restrooms

Preliminary layout:

- Approximate space of 2,50 × 3,20 m;
- Three cubicles on one side;
- Two sinks on the other;
- One exterior door preserved;
- Another door annulled;
- Drainage subject to checking;
- Cubicle doors preferably outward;
- Approximate cubicle depth of 1,20 m;
- Widths subject to partition thicknesses and regulations.

Cubicle systems to be compared:

- Reinforced moisture-resistant board;
- Compact phenolic panel;
- Lightweight rendered masonry.

There is no universally better solution. Cost, space, impacts, humidity, maintenance, and finish must be compared.

11.5. Suspended Ceiling

Criteria:

- Clean dry or with low humidity;
- Test on an inconspicuous panel;
- Do not use steam without verifying compatibility;
- Replace panels that do not recover uniformity;

- Verify format and availability;
- Separate external cleaning from internal cleaning of ducts;
- Do not attribute black stains to a definitive cause solely based on a photo.

11.6. Technical Client

A technical buyer expects:

- measurements;
- unit limits;
- defined materials;
- procedures for authorizing extras;
- documentary review;
- accurate sums;
- prudent timelines;
- delimited guarantees;
- payment form tied to milestones;
- version control.

A high price may be justified if the scope is explained. A price without measurements or vague items generates distrust.

11.7. Organization

Team by fronts:

- bathrooms;
- redistribution;
- painting;
- direction and coordination separated from production.

It is not advisable to count the coordinator as a productive worker if their actual function is:

- procurement;
- quality;
- suppliers;
- incidents;
- client;
- documentation;

- sequences;
- delivery.

12. Case G — Cave or old cellar: stabilization with lime mortar and mesh

12.1. Context and real objective

Case of a cave or auxiliary cellar formed by several small rooms, vaults, natural arches, corners, and very irregular substrates. Old white lime, brown earth or limestone, flaking, surface cracks, built-in masonry benches, and elements typical of a space used as a storage or cellar were observed.

The objective was not to convert it into a smooth room or residential space. It was:

- reduce detachment;
- stabilize ceilings and loose areas;
- prevent continuous falling of earth or finishes;
- preserve the rustic geometry;
- maintain a system compatible with an old substrate that needs to evacuate moisture;
- execute a solution proportionate to the ancillary use of the space.

Working hypothesis used in the historical budget:

- approximate floor area: 35 m²;
- walls and vaults to treat: up to 95 m²;
- galvanized mesh planned: up to 50 m².

These figures were budget limits, not a definitive geometric measurement.

12.2. Air Lime, NHL and Hybrid Mortars

Several materials were compared:

- **Air lime**: highly breathable and traditional, but slow-setting and less suitable for thick applications over damp or weak substrates;
- **Natural hydraulic lime NHL 2 or NHL 3.5**: technically the most compatible option with stone, earth, and ancient buildings; sets with ambient humidity and maintains moderate stiffness;
- **Premixed lime mortar**: facilitates dosing and regularity, but its data sheet must be reviewed;
- **Hybrid lime mortar with hydraulic or cementitious binder**: cheaper and more practical, although typically stiffer and less traditional;
- **Pure cement**: discarded as a general render for the cave; may be reserved for localized brickwork or reinforcements if the system requires it.

In the case, a commercial white render with lime, vapor-permeable and spray-applied, had been tested. It was considered usable on sound areas, but not equivalent to pure NHL restoration mortar.

Editorial Rule:

Do not classify a mortar as suitable merely because the word “lime” appears on the bag. The mortar must be reviewed for binders, strength, thickness, support, and breathability.

12.3. First Hand-applied Coat and Preparation

An experienced worker noted that the first sprayed layer would not adhere well to earth or loose limestone. The final criterion was:

1. Move objects and protect the area;
2. Scrape only loose material;
3. Brush and vacuum loose dust and earth;
4. Do not destroy the remaining firm white lime;
5. Create a small pilot section;
6. Place mesh on ceilings and weak points;
7. Apply the first coat of lime mortar by hand, pressing it against the support and mesh;
8. Leave a rustic finish, without screeding the entire irregular geometry flat.

The pilot section serves to check:

- Adhesion;
- Actual consumption;
- Thickness;
- Work speed;
- Support behavior;
- Appearance accepted by the client.

12.4. Galvanized Mesh

The preferred mesh was triple-twisted galvanized, fine and flexible, more adaptable to vaults than a rigid linked mesh.

Reference studied:

- Approximate mesh spacing of 13 mm;
- Approximate wire diameter of 0,7 mm;
- Roll of around 1 × 50 m.

Placement criteria:

- Anchor to firm stone or limestone, not to loose earth;
- Use screws, anchors, and wide washers;
- Fix more densely on ceilings than on walls;
- Overlap panels approximately 10 cm;
- Do not press the mesh completely against the wall;
- Leave it separated approximately 3–5 mm to allow the mortar to enclose it;
- Apply moderate and well-pressed layers.

Mesh is not prescribed automatically across the entire surface. It is especially useful in:

- Weak ceilings;
- Support changes;
- Earthy areas;
- Joints;
- Edges;
- Non-structural surface cracks.

The mesh does not convert an unstable support into a safe structure.

12.5. Thickness and Consumption

The target thickness ended up being around 10 mm total, applied in one or two passes depending on support. Do not promise a uniform layer of 20 mm across an irregular cave without measuring consumption, added weight, and substrate condition.

On surfaces of this type, consumption depends more on:

- voids;
- roughness;
- material losses;
- areas requiring filling;
- percentage of surface with mesh;
- rustic or regularized finish;

than on square meters of floor.

12.6. Brick Arches and Centring

Adding two brick arches was considered and a small reinforcement wall. The conclusion was that the arches were the part that most increased difficulty and cost.

An arch requires:

- setting out width, height, springing points and curve;
- solid supports;
- timber centring, boards or another sufficiently rigid form;
- placement of bricks with radial joints;
- closure via the keystone;
- curing time;
- dismantling of the formwork;
- finishes and integration with the cave.

An arch should not be built freehand. Sacks or polystyrene may help as fill or secondary form, but they do not replace a stable formwork.

The correct commercial reduction was to eliminate the arches, keeping only stabilization work and necessary reinforcement. It is a clear example of **reducing scope**, not of executing the same work for less money.

12.7. Historical Budgets of the Case

Internal figures from July 2026, useful only to explain price formation:

- option with arches and greater scope: 4.800 €;
- reduced option without arches: 3.950 €.

The 4.800 € budget contemplated approximately:

- 110 bags of prepared lime mortar;
- 10 bags of localized masonry mortar;
- a roll of mesh of 50 m²;
- anchors;
- 180 bricks;
- formwork and battens;
- protections and consumables;
- transportation;
- about 15 labor days;
- organization, finishes, and unforeseen costs.

These figures should not be published as current rates. The editorial value lies in showing that arches, mesh, transportation, manual work, and irregular support change the budget.

12.8. Warranty and Limits

A reasonable warranty may cover detachment or defects attributable to execution within the agreed scope. It must exclude:

- structural movement;
- active water ingress;
- erosion or degradation of the original support;
- detachment outside treated areas;
- behavior of previous materials not removed;
- impacts, use, or subsequent interventions.

It should not be guaranteed that an internal finish permanently eliminates damp from a cave.

13. Case H — Two Adjacent Bathrooms in an Old and Narrow House

13.1. Context

Case of execution of two small, adjacent bathrooms linked to rentable rooms in an old house with a sloped roof, lime and stone walls, wooden beams, and very limited space.

Measures managed during the process:

- interior widths between approximately 1,07 and 1,21 m;
- a critical zone close to 0,99 m before finishes;
- depth close to 2,50 m;
- elevated shower trays to achieve slope;
- tightly constrained drainage between toilets, showers, and partitions.

The objective was robust functionality, not luxury or universal accessibility. Priorities were:

- drainage without counter-slopes;
- inspection access at critical points;
- stable shower bases;
- localized waterproofing;
- flat surface for tiling;
- breathability in old dry areas;
- compatibility with partially embedded beams;
- future repair capacity.

13.2. 90 and 110 mm Drainage

The main collector was PVC of 110 mm. Toilets had a horizontal exit close to 90 mm. The general rule is to transition to 110 mm as soon as possible, but the case showed a practical exception:

A forced, tensioned, and inaccessible 110 mm solution may be worse than a short 90 mm segment, well supported, with slope and inspection, which later transitions to 110 mm.

The critical configuration used two 90 mm branches for the WC before transitioning to 110 mm. It was recognized as a compromise, not an ideal universal solution.

Minimum conditions to accept a similar compromise:

- short segment;
- continuous slope;
- absence of counter-slope;
- pieces without tension;
- mechanical support;
- access for disassembly or unclogging;
- verification by a responsible plumber;
- not automatically applying this solution to another project.

13.3. Fittings, Clearances, and Chemical Welding

Compression couplings may facilitate disassembly, but only when they remain accessible and without stress.

Rules:

- accessible fitting: may be reasonable;
- buried fitting under a tray without inspection: must be avoided;
- do not leave tubes hanging from the fitting;
- use clamps or supports;
- do not extend 90 mm segments only to gain disassembly;
- do not use PVC glue as filler for large clearances.

PVC glue performs a chemical welding between compatible pieces. It does not replace an incorrect socket or resolve diameter differences of 0,5–1 mm. When pieces do not fit, the correct solution is to cut, replace, or seek a compatible adapter.

13.4. Service Enclosures and Inspection Access

Two concepts were separated:

- **fixed and waterproof zone**: simple pipes and low-risk lines that can remain closed;

- **accessible critical zone**: branches, couplings, unions, and points that may require intervention.

Preferences:

- vertical inspection access rather than a horizontal cover in a wet zone;
- sacrificial cover only when it remains out of direct water and its future opening up is accepted;
- do not support hollow clay boards or partitions on pipes;
- the enclosure must not serve as the wall's structure;
- if aesthetics allow, it is better to leave a technical point visible or cover it with a removable panel than to bury it irreversibly.

13.5. Elevated Bases for Shower Trays

Shower trays needed to be elevated approximately 7–8 cm to resolve drainage. Adding the thickness of the tray, the total step could reach around 11 cm.

An additional front edge was discarded because it would increase:

- risk of tripping;
- joints;
- cleaning;
- masonry work;
- delicate points of waterproofing.

The base must offer distributed support, not a hollow frame supported only on edges.

Recommended sequence:

1. clean and vacuum;
2. mark levels with a laser;
3. dry-fit the shower waste and check the slope;
4. moisten the support;
5. apply a bonding agent or cement slurry when appropriate;
6. set reference levels and screed guides;
7. place well-bedded hollow clay boards or blocks to reduce weight;
8. fill solid and form a continuous mortar bed;
9. leave the base flat and level;
10. cure;
11. waterproof;

12. adhere the tray with a compatible product.

The slope is incorporated in the tray; the base must not be arbitrarily inclined.

13.6. Screed and Waterproofing

For screed thicknesses of 5–6 cm, a semi-dry mortar, compacted and ruled, was considered more appropriate than filling everything with costly self-leveling compound.

The studied waterproofing system contemplated:

- full shower;
- full floor of both bathrooms;
- perimeter rise of approximately 25–30 cm;
- bands or reinforcements at corners;
- reinforcement at pipe transitions and support changes;
- localized protection behind sinks.

Indiscriminate waterproofing of all old lime and wood walls was not recommended. Outside the direct water zone, blocking vapor transmission may be counterproductive.

An impermeable liquid membrane may temporarily remain exposed, but in high and visible areas, it must be checked if it allows a compatible bathroom paint. In direct water areas, the usual solution is to protect it with ceramic cladding.

13.7. Moisture-resistant Board and Lime Mortar in Dry Zones

For the shower zone, a moisture-resistant board properly installed or a cementitious panel was preferred:

- controlled plane;
- sufficiently close supports;
- continuous waterproofing;
- adequate support for large tiles;
- less dependence on an irregular old wall.

Bending or forcing a panel to gain one or two centimeters is not recommended. The minimum width loss was accepted in exchange for a stable plane.

In the dry half of the WC and sink, visible or finished lime mortar could be preserved because:

- maintains compatibility with old walls;
- allows drying;
- avoids hiding uncertain beams and supports;
- reduces thickness.

It should not receive direct water or be treated as a shower without protection.

13.8. Wooden Beams

The beams were partially embedded and it was not possible to treat their entire surface. The criterion was to allow the visible face to continue drying.

Avoid:

- impermeable enamel coatings;
- glossy polyurethane finishes;
- fully wrapping the wood with cement, silicone, or waterproof membrane;
- rigidly sealing the wood-lime joint.

Prefer:

- cleaning and vacuuming;
- checking for dryness and condition;
- open-pore wood stain when applicable;
- ventilation via extractor;
- flexible joint or non-rigid junction.

13.9. Tiling and Adhesives

With large pieces of approximately 30 × 90 cm, horizontal placement was considered more manageable. Orientation alone does not determine durability. Important factors are:

- flat support;
- panel without movement;
- appropriate flexible adhesive;
- adhesive applied to both substrate and tile back;
- full coverage;
- regular joints;
- elastic sealing at corners.

For adhering shower trays, domestic water-based adhesives and sanitary silicone as the primary support were discarded. A MS polymer or compatible polyurethane adhesive-sealer was preferred, applied over a clean, dry, stable, and flat base.

13.10. Errors to Avoid

- sealing PVC with bituminous membrane or heat;

- using PVC glue to fill gaps;
- burying removable fittings without access;
- supporting walls or hollow clay boards on pipes;
- creating a hollow tray base;
- adding an unnecessary rim;
- waterproofing over irregular hollow bricks without regularizing;
- bending the panel in the shower;
- using sanitary silicone as structural adhesive;
- sealing wet beams with waterproof finishes;
- tiling entire lime walls without considering breathability;
- promising absolute watertightness without tests.

13.11. Historical Costing and Labor Valuation

The case highlighted a significant difference between:

- formal turnkey company price;
- economically priced materials purchased directly;
- self-executed or partially self-executed work;
- labor compensated through accommodation or barter;
- real cost including the coordinator's time.

Internal reference figures for 2026 were managed as:

- around €8.000 for a local company adjusted for the two bathrooms;
- €4.500–€7.000 with extensive management and partial self-execution;
- higher economic cost once the value of unbilled time is included.

The editorial lesson is not to publish these figures as rates. It is to explain that inexpensive materials do not eliminate the cost of:

- setting out;
- drainage;
- inspection access;
- waterproofing;
- drying;
- testing;

- errors and redoing;
- coordination;
- personal time.

In agreements of work in exchange for accommodation, both parties must value in writing:

- workdays;
- monthly value of accommodation;
- supplies;
- included tasks;
- review date;
- what happens when the initially agreed work is completed.

14. Case I — Client Screening: Clearly Defined Small Job and Large Project Beyond Capacity

14.1. Installation of Fixtures in a Local Bathroom

A small job was studied that included:

- removing existing furniture, sink, mirror, and lighting;
- installing suspended furniture and new sink;
- retaining the faucet if connections matched;
- placing mirror and lamp;
- installing a high suspended auxiliary cabinet;
- hanging an electric towel rail and connecting it to an existing outlet.

A critical clarification was interpreting "columna" correctly: it was not a structural column or a waste pipe, but a high bathroom cabinet.

Estimated duration:

- 6–8 hours if everything fits;
- 8–10 hours if uncomfortable adjustments appear.

The figure dropped from €650 to €420, €350, and finally €300 as it was confirmed that:

- no water-filled radiator was removed;
- no new electrical installation existed;
- no plumbing modifications were made;
- the furniture was provided by the client;

- the work could be resolved in a single day.

This does not mean €300 is a fixed rate. It demonstrates that the price must decrease when uncertainties and line items disappear.

Recommended conditions:

- small materials included;
- payment at completion for low-value jobs;
- does not include pipe adaptation or new wiring;
- does not include replacement of hidden tiles behind dismantling;
- any incompatibility is communicated beforehand.

14.2. Small Work: Day Rate vs. Abstract Hourly Rate

For local installation work, it may be more useful to calculate:

- half-day or full-day work;
- travel;
- procurement or stockpiling;
- difficulty of drilling and leveling;
- risk of finding tiles or deteriorated connections;
- client's commercial value;
- possible revisions;

rather than automatically multiplying all hours by a high hourly rate.

This does not justify charging below cost. It allows adjusting the price to the real effort and risk involved.

14.3. "Blind" Budget for a Guesthouse or Bar

Documentation was received for a renovation and expansion for tourist and bar use, with an architect's valuation of 37.250 €.

The document included a partial package with:

- footings;
- anchor plates;
- metal columns and beams;
- elevator shaft and pit;
- new staircase and stringers;
- railings;

- openings;
- manholes;
- partitions;
- floor preparation.

It did not represent the cost of the finished building nor necessarily a contractor's offer.

A "blind" budget may be:

1. a measurement without prices for bidders to complete;
2. an architect's approximate estimate;
3. a partial package that does not include all specialties.

It must never be compared to an execution offer without verifying:

- scope;
- VAT;
- margin;
- auxiliary means;
- project and technical supervisor;
- insurance;
- travel;
- unforeseen events;
- what each party supplies.

14.4. Elevator: Civil Work vs. Equipment

The package included the shaft, pit, and structural preparation. There was no evidence that it included:

- cab;
- guides;
- machinery;
- landing doors;
- installation by an elevator company;
- regulatory approval;
- commissioning.

Editorial Rule:

"Preparing the elevator shaft" and "supplying and installing an elevator" are entirely distinct items.

14.5. Filter Before Visiting

The work was located at a significant distance from the base and presented structure, use change, accessibility, fire safety, and elevator requirements. Before traveling, a pre-screening should be conducted:

- call or brief audio;
- confirm whether the request is for full work or items;
- ask who the technician is and if there is a license;
- check if the architect's figure is the budget limit;
- ask about timeline and financing;
- confirm who assumes structure and elevator;
- decide if the work fits the company's real capacity.

The cost of a distant visit includes unproductive hours, vehicle use, and loss of capacity to attend viable projects.

14.6. Business Limitation

The conclusion was not to assume the project as the main contractor due to:

- structural intervention;
- public use;
- elevator;
- high technical responsibility;
- advanced coordination;
- uncertainty between plans and reality;
- initial phase of Riansares 4R.

Independent and very delimited items could be studied, but not assume structure, pit, elevator shaft, or global coordination without team, insurance, and sufficient technical support.

Saying no to a project can be a correct business decision.

Expansion — Usage Conditions and Tests Before Closing the Work (November 2026)

Bathrooms: Pre-tiling Checks

In bathroom renovations, before closing up walls and applying finishes, it is advisable to verify water connections, drainage, support, seals, and waterproofing solutions. A moisture-resistant board does not mean complete waterproofing. In narrow bathrooms, the dimensions, drain height, and access conditions determine the final solution.

Offices and Premises: Scope Subject to Use

The constraints of occupancy, accessibility, ventilation, safety, and continuity of activity depend on the building's function. An example of an office does not authorize or certify the execution of warehouses, bars, or public facilities; these cases are treated as general guidance in guides/R4R_85_OFFICES_COMMERCIAL_PREMISES_AND_OWNERS_ASSOCIATIONS.md.